Powertrain Control HLD

Synthetic AUTOSAR-inspired High-Level Design Knowledge Document

Important: This document is fictional, independently created for academic evaluation, and contains no confidential OEM data or reproduced AUTOSAR standard text.

Field	Value
Document ID	SYN-HLD-PT-001
Version	1.0
Status	Approved synthetic baseline
Classification	Synthetic academic test data
Intended use	RAG, OCR, extraction, comparison, and evaluation
Authority	Not valid for production design or vehicle release

1. Purpose and system context

The synthetic powertrain control architecture coordinates driver demand, vehicle motion, engine state, torque arbitration, and thermal protection. The document is designed to support entity extraction, relationship mapping, cited question answering, and deterministic completeness checks.

1.1 Operational boundary

The logical boundary includes application software components only. Hardware drivers, operating-system internals, calibration implementation, and production safety approval are outside the synthetic model.

2. Software component catalogue

Component ID	Software Component	Responsibility	Provided Interface	Required Interface
SWC-PT-001	EngineControlCoordinator	Coordinates torque and engine operating states	If_EngineState	If_VehicleSpeed, If_PedalPosition
SWC-PT-002	VehicleStateManager	Publishes normalized vehicle speed and motion state	If_VehicleSpeed	If_WheelSpeedRaw
SWC-PT-003	PedalPositionAdapter	Validates accelerator pedal channels	If_PedalPosition	If_AdcPedal
SWC-PT-004	TorqueArbitrator	Combines driver and supervisory torque requests	If_TorqueRequest	If_PedalPosition, If_EngineState
SWC-PT-005	ThermalProtectionManager	Applies torque derating from temperature	If_ThermalLimit	If_CoolantTemperature

3. Interface catalogue

Interface ID	Interface Name	Type	Provider	Consumers	Key Data Elements
IF-001	If_VehicleSpeed	Sender-Receiver	VehicleStateManager	EngineControlCoordinator, CruiseControlManager	VehicleSpeed_kph, MotionState
IF-002	If_PedalPosition	Sender-Receiver	PedalPositionAdapter	EngineControlCoordinator, TorqueArbitrator	PedalPosition_pct, PedalValidity
IF-003	If_EngineState	Sender-Receiver	EngineControlCoordinator	TorqueArbitrator, DiagnosticEventManager	EngineRunning, EngineSpeed_rpm
IF-004	If_TorqueRequest	Client-Server	TorqueArbitrator	EngineControlCoordinator	RequestedTorque_Nm, RequestSource
IF-005	If_ThermalLimit	Sender-Receiver	ThermalProtectionManager	EngineControlCoordinator	MaximumTorque_Nm, DeratingReason

4. Port mapping

Port ID	Component	Port	Direction	Interface	Multiplicity
PORT-001	VehicleStateManager	PpVehicleSpeed	Provided	If_VehicleSpeed	1
PORT-002	EngineControlCoordinator	RpVehicleSpeed	Required	If_VehicleSpeed	1
PORT-003	PedalPositionAdapter	PpPedalPosition	Provided	If_PedalPosition	1
PORT-004	TorqueArbitrator	RpPedalPosition	Required	If_PedalPosition	1
PORT-005	EngineControlCoordinator	PpEngineState	Provided	If_EngineState	1
PORT-006	EngineControlCoordinator	RpTorqueRequest	Required	If_TorqueRequest	1

5. Functional flows

5.1 Driver torque request flow

PedalPositionAdapter validates two synthetic pedal channels and publishes PedalPosition_pct through If_PedalPosition. TorqueArbitrator consumes the value, considers EngineRunning from If_EngineState, and exposes RequestedTorque_Nm through If_TorqueRequest. EngineControlCoordinator consumes the request and applies the permitted torque.

5.2 Vehicle-speed flow

VehicleStateManager receives WheelSpeedRaw, calculates VehicleSpeed_kph, and publishes it every 20 ms through PpVehicleSpeed. EngineControlCoordinator consumes the signal at RpVehicleSpeed for torque coordination.

5.3 Thermal limitation flow

ThermalProtectionManager publishes MaximumTorque_Nm and DeratingReason through If_ThermalLimit. EngineControlCoordinator limits the requested torque whenever the thermal limit is lower than RequestedTorque_Nm.

6. Timing and quality constraints

Item	Requirement
Vehicle speed publication	20 ms nominal; stale after 100 ms
Pedal position publication	10 ms nominal; invalid on channel disagreement
Torque request service	Response within 15 ms
Thermal limit publication	100 ms nominal
Invalid input behaviour	Use last valid value for one cycle, then enter degraded mode

7. Assumptions and limitations

All names and values are synthetic. A matching provided port is expected for every required port. Consumers must not infer relationships that are absent from this document.